

Society of Critical Care Medicine 2024 Guidelines on Adult ICU Design: Executive Summary

KEYWORDS: evidence-based design; guidelines; healthcare design; intensive care unit architecture; intensive care unit design

Advances in technology, challenges in infection control—such as the severe acute respiratory syndrome coronavirus 2 pandemic, and evolutions in patient- and family-centered care highlight ideal aspects of ICU design present opportunities for enhancement (1, 2). For example, prior Society of Critical Care Medicine (SCCM) ICU design guidelines (1995–2012) did not envision remote manipulation of ventilator settings or infusion pumps (3, 4) or the unique aspects of pandemic care. Design elements spanning square footage, air handling, airborne isolation, linkage to electronic and digital local or remote systems, as well as ICU organization and layout may be addressed during new construction, revision of existing critical care spaces, or conversion of previously noncritical care space to render ICU care. Ensuring proximity to key destinations helps enable safe, quality care for all ICU subspecialties. ICU design may influence safety and security for patients, visitors, and staff (5). Due to substantial shifts in healthcare and intervening research, SCCM sought to update the 2012 ICU design guidelines to provide expert guidance for clinicians, administrators, and healthcare architects considering constructing a new ICU or renovating one.

ICU DESIGN POPULATION, INTERVENTION, COMPARISON, AND OUTCOMES QUESTIONS

A summary of Good Practice Statements (GPSs) and Strong Recommendations for selected Population, Intervention, Comparison, and Outcomes (PICO) questions are presented in **Table 1** along with panel generated design themes. PICO questions in “bold italics” represent the most impactful areas determined by the panel and are presented herein. **Figure 1** provides a Visual Summary of the certainty of evidence and strength of recommendations for each PICO question. Evidence summaries and recommendation justifications for all 15 questions are located within the supporting materials. Overall, the panel articulated 17 recommendations (PICO questions 2.1. and 5.2. each yielded two recommendations), including five GPSs.

High Level Summary of PICO Questions With Strong Recommendations and GPS

Theme 1: ICU Layout.

1.1. Should high-visibility vs. low-visibility layouts be used in ICUs? One primary determinant of patient visibility is ICU layout. Visibility of patients at risk of deterioration is a high priority and complements existing monitoring

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TABLE 1.

Complete Summary of ICU Design Themes and Related Population, Intervention, Comparison, and Outcomes Questions^a

Theme	Population, Intervention, Comparison, and Outcomes Question
1. ICU layout	1.1. Should high-visibility layouts vs. low-visibility layouts be used in ICUs? 1.2. Should centralized charting areas vs. decentralized charting areas be used in intensive care? 1.3. Should single-bed rooms vs. open bay layouts be used in ICUs? 1.4. Should designs with close proximity to key destinations vs. without close proximity to key destinations be used for ICUs?
2. Room design	2.1. Should rooms with environmental features to enhance sleep and recovery vs. standard rooms be used in ICUs 2.2. Should in-room supplies vs. centralized supply rooms be used in ICUs?
3. Infection control	3.1. Should advanced HVAC designs vs. standard HVAC designs be used in ICUs? 3.2. Should advanced infection prevention features vs. no advanced infection prevention features be used in ICUs?
4. Infrastructure	4.1. Should outside-room monitoring and control of devices vs. inside-room only monitoring and control of devices be used in ICUs? 4.2. Should advanced remote monitoring (e.g., telemedicine) vs. usual care be used in ICUs? 4.3. Should flexible surge capacity vs. no specific design for surge capacity be used in ICUs? 4.4. Should nonwall-based life support utility access vs. wall-based life support utility access be used in ICUs?
5. Staff space	5.1. Should ergonomic features vs. usual designs be used for ICUs? 5.2. Should integrated break/respice space vs. nonintegrated break/respice spaces be used in ICUs? 5.3. Should mobile workstations, or combination workstations vs. fixed workstations, be used in ICUs?

HVAC = heating, ventilation, and air conditioning.

^aItems in “bold italics” represent the one Population, Intervention, Comparison, and Outcomes question in each theme that have been selected for presentation within the Executive Summary; all others are fully reviewed in the complete article (6).

devices, as the sickest patients benefit from early problem detection (7–10). Caring for patients in more visible areas may allow staff to more rapidly intervene and to recognize when colleagues require assistance. The panel noted that “visibility” specifically refers to the patient including their face, monitors, and bedside alarms—as opposed to the room entryway or nonpatient-care design elements.

A Strong Recommendation was made in favor of high visibility, despite a low certainty of evidence that evaluated patient safety during critical illness. Although the certainty of evidence is low, this is a fundamental aspect of ICU care. The undesirable effects of high-visibility rooms (e.g., reduced privacy) are believed to be minimal by comparison to the anticipated benefits and may be easily mitigated (11). ICU design for optimum patient visibility from staff workstations is a priority.

Theme 2. Room Design.

2.1. Should rooms with environmental features that enhance sleep and recovery (light and noise mitigation, natural lighting) vs. standard rooms be used in ICUs? These aspects are priorities as ICU environments commonly disrupt natural sleep cycles, promote delirium, and impede recovery. Incorporation of natural lighting, dynamic lighting, and noise mitigation could reduce sleep disruption. While early studies of windows suggested an impact upon mortality and delirium, effects remain unclear. Due to confounding risks in observational studies, as well as effect estimate imprecision, the panel assigned a low certainty of evidence for window and natural lighting effects on mortality, delirium, as well as ventilator or ICU length of stay. Windows are inherently desirable as they humanize the critical care setting, reflect current patient, family,

SCCM ICU Design Guideline - Visual Summary

Theme 1: ICU Layout



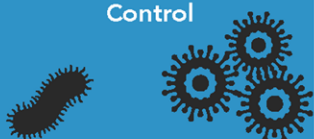
- ✓✓ High visibility layout ⊕⊕○○
- ? Centralized or decentralized ICU charting ⊕⊕○○
- ✓ Single patient rooms over open-bay layout ⊕○○○
- 👉 Close proximity to critical hospital areas

Theme 2: Room Design



- ✓✓ Windows and natural lighting in all patient rooms ⊕⊕○○
- ✓ Noise reduction strategies ⊕○○○
- ? In-room and/or centralized supply areas ⊕○○○

Theme 3: Infection Control



- ? Standard or advanced HVAC systems ⊕○○○
- 👉 Features to prevent airborne, water-borne, and surface transmission

Theme 4: Infrastructure



- ✓ Capacity for controlling devices outside patient rooms ⊕○○○
- ✓ Tele-medicine capacity ⊕○○○
- ✓ Non wall-based life support utilities access ⊕○○○
- 👉 Plan for surges in patient volumes

Theme 5: Staff Space



- ✓ Ergonomic designs in workspaces and patient care areas ⊕⊕○○
- ✓ Access to quiet rooms and respite spaces ⊕⊕○○
- 👉 Dedicated break rooms
- 👉 Accommodate fixed and mobile workstations

Legend

STRENGTH OF RECOMMENDATION

- ✓✓ Strong Recommendation
- ✓ Conditional Recommendation
- ? No recommendation
- 👉 Best Practice Statement

CERTAINTY OF EVIDENCE (GRADE)

- ⊕⊕⊕⊕ Very high
- ⊕⊕⊕○ Moderate
- ⊕⊕○○ Low
- ⊕○○○ Very low

Figure 1. Society of Critical Care Medicine (SCCM) ICU design guidelines, all Population, Intervention, Comparison, and Outcomes—visual summary. GRADE = Grading of Recommendations, Assessment, Development and Evaluation, HVAC = heating, ventilation, and air conditioning.

and staff expectations and are encoded in existing ICU standards. A strong recommendation was made supporting windows in patient rooms.

Studies of specific design-related features to address ICU noise mitigation were not identified. Noise canceling ceiling tiles may enhance patient rest and staff communication (12). Common ICU noise sources include staff activity and conversation, furniture movement, other patients, visitors, and device alarms. Because alarms often exceed the World Health Organization decibel standards, they are associated with impaired sleep hygiene (13–16). The panel agreed that the effect of ICU design noise mitigation strategies warranted a very low certainty of evidence assessment due to limited study data.

Theme 3. Infection Control.

3.2. Should advanced infection prevention features vs. no advanced infection prevention features be used in ICUs? Nosocomial infection is a challenging source of morbidity and mortality in the ICUs and localized outbreaks are well described (17). There is no strong evidence supporting the efficacy of any single infection prevention/control measure to address nosocomial infection. Many measures may reduce microbe prevalence on surfaces, in air, and in water. It is less clear that these measures result in reduced colonization and subsequent infection, but they offer interventions designed to reduce the likelihood of nosocomial pathogen acquisition and subsequent infection, especially in those with immune compromise. Studied interventions included: 1) reducing or clearing pathogen bioburden (18–20); 2) improving hand hygiene compliance (21–25); 3) concerns regarding sink location, splash guard use, and water filter emplacement (26–33); 4) appropriate space for personal protective equipment storage and use (34); 5) pathogen-reducing or surface-cleaning enabling surface materials (35–46); and 6) the impact of push-plate door handles (47). Most interventions demonstrate face validity and appear to reduce microbe counts on surfaces as well as patient colonization by antimicrobial-resistant or multidrug-resistant organism pathogens.

While it is unclear which single advanced infection prevention and control feature is most effective, the cumulative effect of multiple simultaneous interventions to mitigate nosocomial colonization, infection, and localized outbreaks is anticipated to be large. A GPS recommendation to incorporate design

features to prevent airborne, water-borne, and surface transmission.

Theme 4. Infrastructure.

4.3. Should flexible surge capacity vs. no specific design for surge capacity be used in ICUs? The COVID-19 pandemic highlighted the unpredictability of critical care needs and the importance of being able to rapidly augment bed capacity to address patient volume surges. Surge capacity includes equipment, staff, and the ICU physical infrastructure (i.e., beds or care locations). While comparison studies of surge capacity were not identified, strategies to rapidly increase capacity included: 1) cohorting multiple patients within a single room (48, 49); 2) using novel spaces for patient care (50, 51); 3) leveraging resources across health systems such as load balancing across sites (52); 4) deploying infant monitors to increase observation capability (50, 53); and 5) emplacing portable high-efficiency particulate air filters to improve airborne isolation room complement (50). Designs that accommodate large patient volume surges may support continued access to routine as well as emergency care despite system stress. Additionally, staff augmentation may occur using a tiered-staffing structure where ICU clinicians guide teams of non-ICU clinicians to provide critical care during surges (54).

Theme 5. Staff Space.

5.2. Should integrated break/respice space vs. nonintegrated break/respice spaces be used in ICUs? Staff satisfaction, burnout, and clinical performance may be influenced by the design, usability, and impact provided by nonworkspaces such as break rooms and respice areas. Break rooms are often multifunctional, providing space for nourishment, team education, as well as team bonding and mentoring. Such spaces may promote staff well-being. Since critical care environments are often high-stress environments, individual spaces devoted to recovery and well-being complement breakroom functionality.

The panel made two recommendations. First, including dedicated staff break rooms that provided storage lockers, washrooms with showers, and nutrition areas was embraced as a GPS. An additional consideration is to locate the break room within the ICU, in a space with windows for natural light. Second, a conditional recommendation was crafted for less essential “wellness rooms” or “respice spaces” as promising complements to break rooms, noting that there is limited evidence to support this as a routine practice (55, 56).

CONCLUSIONS

This executive summary and associated article are SCCM evidence-based guidelines, including 15 PICO questions that update SCCMs 2012 guidelines. The guidelines panel considered five themes—layout of ICU rooms, room design, infrastructure, infection control and prevention, and space for staff—as domains related to ICU design. This summary presents five of the 17 recommendations that if implemented will result in ICU designs that are patient, family, and clinician centered. Strong Recommendations were made for: 1) high patient visibility and 2) room environmental features that enhance sleep and recovery. Other recommendations were conditional along with GPSs including: 1) integrated staff break/respite spaces, 2) advanced infection prevention features, and 3) flexible surge capacity design. While the underpinning evidence was of low certainty, these guidelines provides a unique and comprehensive summary of evidence-based design data informed by practice-based expertise.

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The Society of Critical Care Medicine guidelines are intended for general information only, are not medical advice, and do not replace medical professional advice, which should be sought for any medical condition. The full disclaimer for guidelines can be accessed at: <https://sccm.org/Clinical-Resources/Guidelines/Guidelines>.

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